

VECTOR — Current Research State

Last synced: 2026-09-24
Scope: Concise research dashboard; detailed evidence and history remain in the linked sources. This dashboard distinguishes source review from reproduced empirical results.

Current scientific question

How do assortative assignment, congestion, and selection scarcity K/N interact? Does assignment preference create, amplify, or reshape congestion's effect on advancement? Can congestion change outcomes without assortative preference, and could extreme scarcity make men's basketball (MBB) a boundary regime? Similar outcome curves may arise from different mechanisms. These are open questions. [S1–S2]

Current advisor priorities

The latest advisor discussion represented in the reviewed brief is PD41 (September 22): resolve whether assortativity matters before reshaping the presentation. PD40 explicitly allows the answer that assortativity is unnecessary and cautions against generalizing from basketball alone. PD41 also recommends a bounded education review, specifically checking college destination/prestige, before presentation restructuring. VECTOR has now read PD41 and PD30 directly and inspected the relevant PD40 passages. Charles reaffirmed assortativity as the main focus and requested the education review on September 24. [S1; S2 §1; S5]

Established evidence and present baseline

- **Binding framework:** distinguish ASSIGN, SCORE, and SELECT; an observed outcome curve does not identify its mechanism. The binding document separately distinguishes peer environment from advancement. [S2 §2; S3]
- **Directly inspected documentation:** the working brief specifies an eight-cell comparison crossing ρ , λ , and K/N . The design exists; this experiment has not been executed. [S2 §§3–4]
- **Saved, unreproduced output:** the brief records a Grandchild sorting sweep with positive realized sorting at $\rho = 0$. This supports keeping assignment preference separate from realized sorting; it is not a result for the proposed factorial population. [S2 §5, S11 therein]
- **Army status:** the September 24 handoff reports Runs 1–3 completed, Runs 4–5 pending, and a planned own-zero plot gate. These are reported progress, not independently verified refreshed figures; the refresh remains **PARTIAL**. [S2 §6b; S4]

Education: both supplied panels contain entry and completion indicators but no college destination/selectivity field. The three inspected saved mosaics show generally rising attainment with peer quality, including within high-ability bands. These are saved descriptive results, not newly reproduced estimates. [S5]
No additional reproduced empirical finding is established by this dashboard.

Active hypotheses

Extreme scarcity may permit congestion effects without assortative preference. Assortativity may instead amplify or reshape effects, and different parameter combinations may produce similar curves. Army's band-of-excellence framing is a complementary mechanism hypothesis. A proposed education lens separates school standing, postsecondary entry, destination, and completion; any competition at the access stage remains a hypothesis. These questions are under design and evidence review, not experimental testing. [S1–S2; S5]

Unresolved contradictions

- **Basketball:** calibration versus replay probabilities and parameter transfer; annual versus ever-drafted outcomes; empirical versus simulated axes; local tail decline versus global quadratic curvature.
- **Army:** straight-LOO versus earlier peer definitions; $H_{\text{sort}} = 0.131$ versus approximately 0.255–0.279; Mac/AWS run-to-result mapping; completed components versus unfinished mosaic.
- **Tenure:** construction roster versus inference sample versus decision cohort; person-year versus decision-year analyses; completed versus unfinished survival work.
- **Education:** original outcome recodes and extraction provenance need verification; college identifiers/selectivity and admissions records are absent from the current extracts. Sampled school standing does not directly measure the college applicant pool or college peer environment. [S5]
- **Terminology:** historical Pass A/B/C and "Layer B" labels must retain their document-specific meanings.

The earlier reconciliation questions remain open; education adds source-definition and availability gaps identified in the current review. No scientific discrepancy is treated as resolved here. [S2 §7; S5]

Current task

Charles is reading the PD41 brief. VECTOR has completed a bounded education source review and prepared a transitions-based lens for discussion. Remain in scientific-design and evidence-analysis mode; education does not displace the main assortativity question or resolve the calibration/replay limitation. [S1; S5]
Charles reports that a faculty colleague has source access. At his request, the education review now includes a targeted data-request addendum, prioritizing NELS destinations/selectivity, applications/decisions, actual school rank, and extraction documentation; HS&B is second. Available fields and the permitted working arrangement remain unverified. No new data have been received or analyzed. [S5, addendum]

Decisions made

Charles authorized routine maintenance of VECTOR-owned research documents in this workspace and designated the brief a working document. Substantive scientific changes must be explained before adoption as authoritative. Analytical execution and Git operations require separate explicit authorization. The source operating rules govern the full scope. [S1]

Immediate next decision/action

Discuss the brief’s proposed exact, unclipped top-*K* rule; fixed viability and scaling assumptions; scenario values; tolerances; and paired replication plan. Before requesting execution, specify the exact files, held-fixed quantities, manipulations, outcome interpretations, and provenance risks. For education, discuss a source/coding audit followed by a pre-specified entry-versus-completion comparison; destination selectivity requires a separate availability check. No execution approval is recorded for either investigation. [S1 §7; S2 §8; S5]

Sources

- **S1:** [Charles’s standing repository authority](#).
- **S2:** [PD41 working scientific brief](#).
- **S3:** [Binding separation of environment, scoring, and selection](#).
- **S4:** [September 24 Army handoff](#).
- **S5:** [Education transitions and congestion review](#).